COIT12209

LECTURE NOTES

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1 What is Data Science?

1.1 Data Science:

The collection, preperation, analysis, visualization, management and presentation of large collections of data. Data science specifically is the exploration of data via the scientific method. It is crucial we understnad data science as we are in an age where everything is going digital. For example, targeted advertising and sponsored results.

1.2 History of Data Science:

FAANGY (Facebook, Apple, Amazon, Netflix, Google and Yahoo!) realised they have copious amount of data laying around. If they applied machine learning techniques to the data, significant business benefits occur. These can include better product recommendations, fraudulant activity monitoring, and click through rates.

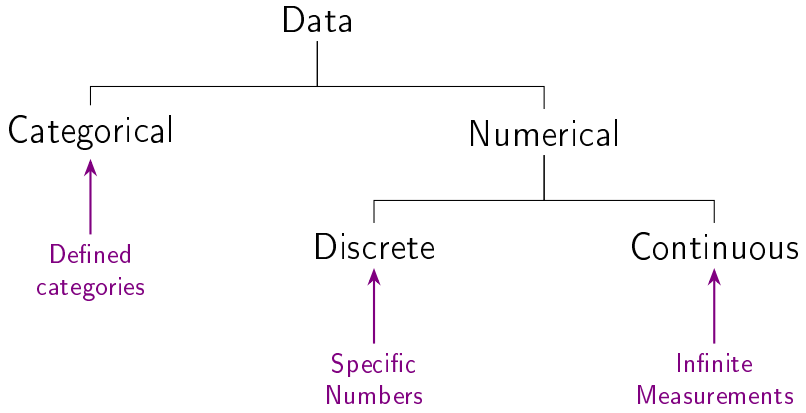
1.3 Data Science Sub-Fields:

- Mathematics and Statistics
- Machine Learning
- Software Engineering
- Domain Knowledge

1.4 What is Data?:

Data usually refers to raw/unprocessed facts. Once anything is done to these facts, in way normies can understand, these facts are considered information.

1.4.1 Data Types



1.5 Big Data:

Comprised of 3 main characteristics:

1. Volume
2. Velocity
3. Variety

Structured Big Data is the common. The formats this comes in is SQL, or CSVs. The positives of these formats is that they can be easily searched. However, must follow strict schema, offering limited flexibility.
Semi-structured have a litt bit of organization like tags or keys, but does not follow strict schema rules. This giver more flexibility too the data but makes it harder to query. Formats can include XML, YAML, and JSON.
Unstructured data follows no consistent formatting. These include text files, PDFs and A/V files. They are rich in formation but hard to store and query efficiently.
Big Data plays a crucial role in industry as it allows data driven descion making.

2 Identifying Data

2.1 Data Mining Task:

Data mining tasks are specific operations or processes to find extra patterns or insights from the data. By breaking down big data into smaller chunks an easier

insight can be found. This allows humans to only do the part where creativity can be found, I.E engaging with stakeholders.

2.2 Business Problems:

Business analystics and problem solving require data scientists to find "correlations" between variables, their meanings and relationships to other variables. In additon it allows informed descion making based on a data driven approach.

2.3 Data Mining Task:

Data mining taks are designed to extract insights or knowlegde from a data set. This is split into 2 categories:

1. Predictive tasks: Use historic data to predict future values
2. Descriptive tasks: Discovering patterns without prior labels

2.3.1 Predictive Tasks

- Classification and class probability: Predicting a category label for each instance.
- Regression: Predict a continuous numeric value.
- Similarity matching: Predict which past records are most alike a new input.
- Link prediction: Predicting future or missing connections in a network.
- Casual modeling: Estimation of the effect of an invention.

2.3.2 Descriptive Tasks

- Clustering: Discovering groupings or clustering within data sets.
- Co-occurence grouping: Items found frequently together.
- Profiling (Behaviour description): Describes typical behaviour within a group.
- Data reduction: Simplify or compress data while preserving key information.

2.4 Classification:

Predicting discrete class labels from the class probability. That being the likelihood that an instance belongs to a particular class.

2.5 Data Collection:

1. Manual Collection
    - Observing and recording
    - Interviews
    - Questionares
    - Pros:
        - Fine-tuned
        - Accurate
    - Cons:
        - Scale
        - Cost
        - Reachablility
2. Automatic Approach
    - Transaction history or logs
    - Sensors suchas, GPS, IoT or Lab equipment
    - Web crawler
    - Social media
    - Pros:
        - Mass collection
        - Easy
        - Cheap on a large scale
    - Cons:
        - Mass collection comes with increased mess

2.6 Sampling:

Data sampling is a statistical analysis technique used to select, manipulate and analyse a subset of representative data. This is done to reduce compute time, resource allocation, and time limits. Sampling types include:

- Simple random sampling
- Stratified sampling
- Cluster sampling
- Systematic sampling
- Multi-stage sampling

2.7 Data Mining Process:

Industry standard is CRIP-DM. This stands for Cross Industry Standard Process for Data Mining.

3 Ethics and Law

3.1 Ethics, Morality and Law:

Ethics is purely a branch of philosophy that consist of a set of values to define the rightness and wrongness of an action. Morality is th ethics and belief held by a group, and usually constituated from cultural and relgious principles. Law is defined by a set of rules determined by a governing body. Usually informed through legal instituation and carries consequence to those who break it.

3.2 Organizational Evil:

OE involves systemic or institutional wrong doing that harms consumers. Basically designed to maximise a companies profits. OE can take various forms:

- Unethical practices
- Abusive behaviour
- Corruption
- Disegard to human-rights.

3.3 Digital Era Bias:

Bias is shown through favouritism toward opinions, action or descions.

1. Algorithmic Bias: AI powered systems ignore bias training data to favour particular outcomes.
2. Sampling Bias: Showed dataset containing societal inequalities. Usually happens because of wrong sampling processes.
3. Confirmation Bias: Recommendations enforcing existing beliefs.
4. Representation Bias: Occurs when certain groups or characteristics are over represented.
5. Accessibility Bias: Design choices affecting those with a disability.
6. Language Bias: LLMs may exhibt a bias towards a particular language.

3.4 Gen AI Ethics:

Gen AI should address concerns related to its generated content.

3.5 Ethical Dilemma:

This occurs when tow or more ethical principles conflict. A solution can be difficult to find.

3.6 Ethical Data Science:

EDS refers to a set of moral principles and guidelines outlining data use. Its about making companies think about utilizing data fairly. Australia subscribes to the follow 10 rules for data research:

1. Acknowledge that data are people and can do harm.
2. Recognize that privacy is more than a binary value.
3. Guard against the reidentification of your data.
4. Practice ethical data sharing.
5. Consider the strengths and limitations of your data; big data does not automatically mean better.
6. Debate the tough, ethical choices.

7. Develop a Code of Conduct for your organization, research community, or industry.
8. Design your data and systems for auditability.
9. Engage with the broader consequences of data and analysis practices.
10. Know when to break these rules.

4 Data Presentation

4.1 Data and Computers:

Computers deal with many different types of data when processing commands and presenting answers. These can be text, audio, video or pictures.

4.2 Analog and Digital:

Information is presented in 2 main ways:

1. Analog: A continuous representation that changes smoothly over time.
2. Digital: Discrete values, that are broken up using binary code.